TOPIC 02 · CONFIDENCE INTERVALS

Confidence Intervals (z-Intervals & t-Intervals)

Universal interval structure, critical values (z^* vs. t^*), standard error formulas, minimum sample size calculations, and exam-safe interpretation templates.

1. Universal Structure & Variable Definitions

GENERAL CONFIDENCE INTERVAL FORMULA

$$\text{Confidence Interval} = (\text{Point Estimate}) \pm (\text{Critical Value}) \times (\text{Standard Error of Statistic})$$

$$\text{CI} = \text{PE} \pm \text{ME} = \text{PE} \pm CV \cdot \text{SE}$$

Width of Interval: $\text{Width} = \text{Upper} - \text{Lower} = 2 \cdot \text{ME}$. Half the width gives the Margin of Error: $\text{ME} = (\text{Upper} - \text{Lower})/2$, and midpoint gives $\text{PE} = (\text{Upper} + \text{Lower})/2$.

SYMBOL	NAME	DEFINITION & NOTES
PE	Point Estimate	Best single-value guess of the target parameter ($\hat{p}$, $\bar{x}$, $\hat{p}_1 - \hat{p}_2$, or $\bar{x}_1 - \bar{x}_2$). Always lies at the exact center of the interval.
***C*, α**	Confidence Level & Tail Area	C = middle area (e.g., 0.90, 0.95, 0.99); $\alpha = 1 - C$ = total area in both tails ($\alpha/2$ per tail).
z^*	Normal Critical Value	Standard Normal quantile cutting off upper area $(1 - C)/2$. Common values: 90% → **1.645**, 95% → **1.960**, 99% → **2.576**.
t^*	Student's t Critical Value	Critical value from $t(df)$ distribution; always larger than z^* to account for estimating σ with s.
df	Degrees of Freedom	One-sample / paired t: $df = n - 1$. Two-sample t: Welch's formula (from tech) or conservative $\min(n_1 - 1, n_2 - 1)$.
ME	Margin of Error	$\text{ME} = CV \cdot \text{SE}$. Measures maximum expected estimation error at confidence level C (does *not* include bias/nonresponse error).

2. All Standard Confidence Interval Formulas

PROCEDURE	TARGET PARAMETER	FORMULA	NORMAL / LARGE CONDITION
1-Prop *z*-Interval	p	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$
2-Prop *z*-Interval	$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	All 4 counts $n_1\hat{p}_1$, $n_1(1-\hat{p}_1)$, $n_2\hat{p}_2$, $n_2(1-\hat{p}_2) \geq 10$
1-Sample *z*-Interval (σ known — rare)	μ	$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$	Population Normal OR $n \geq 30$
1-Sample *t*-Interval (σ unknown)	μ	$\bar{x} \pm t^*_{n-1} \frac{s}{\sqrt{n}}$	Pop. Normal, OR $n \geq 30$, OR no strong skew/outliers
Paired *t*-Interval (Differences $d_i = x_{1i} - x_{2i}$)	μ_d	$\bar{d} \pm t^*_{n-1} \frac{s_d}{\sqrt{n}}$	Check Normal/Large on the *list of differences* (n = pairs)
2-Sample *t*-Interval (Independent groups)	$\mu_1 - \mu_2$	$(\bar{x}_1 - \bar{x}_2) \pm t^*_{df} \sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}$	Both groups meet Normal/Large condition (n_1, $n_2 \geq 30$ or plots OK)

3. Solving for Minimum Sample Size (n)

To guarantee a Margin of Error at most ME at confidence level C, solve $z^* \cdot \text{SE} \leq \text{ME}$ for n and **always round UP** to the next whole integer.

MINIMUM SAMPLE SIZE FORMULAS (ALWAYS ROUND UP!)

For Proportion p: $n \geq p^*(1-p^*)\left(\frac{z^*}{\text{ME}}\right)^2$ | **For Mean μ:** $n \geq \left(\frac{z^* \cdot \sigma}{\text{ME}}\right)^2$

Variables: p^* = prior guess for p (if no prior estimate is given, use conservative worst-case $\boldsymbol{p^* = 0.5}$ because $0.5(0.5) = 0.25$ maximizes variance); σ = known/planning population SD.

4. What Affects Margin of Error & Interpretation Rules

CORRECT INTERPRETATION TEMPLATES

1. Interpreting the Interval:
"We are *C*% confident that the interval from [Lower] to [Upper] captures the **true [population parameter in context]**."

2. Interpreting Confidence Level (*C*%):
"If we took many random samples of the same size *n* and built a *C*% CI from each, about *C*% of those intervals would capture the true [parameter]."

HIGH-FREQUENCY EXAM TRAPS

- **Never say "95% probability"** that the true parameter is in a *specific* calculated interval! Once computed, the parameter is fixed, so it is either in (1) or out (0).
- **Higher C → Wider interval** (z^* increases). **Larger n → Narrower interval** (SE shrinks by $1/\sqrt{n}$).
- **2-Prop CI never uses pooled $\hat{p}_c$!** Use individual $\hat{p}_1$ and $\hat{p}_2$ (pooled $\hat{p}_c$ is only for 2-Prop *z-Tests*).

USING A CONFIDENCE INTERVAL TO TEST A TWO-SIDED HYPOTHESIS

A two-sided C% confidence interval corresponds to a two-sided significance test at $\alpha = 1 - C$. If the null value (p_0, μ_0, or 0 for a difference) falls **outside** the CI, reject H_0 at level α; if it falls **inside**, fail to reject H_0.